Question 1 — P3   Target 500ml   12 sub $n=5$

(a)
$$\bar{X} = \frac{\sum \bar{X}}{12} = \frac{6011.7}{12}$$

$$\bar{\bar{X}} = 500.975\text{ml}$$

$$\bar{R} = \frac{\sum R}{12} = \frac{57.2}{12}$$

$$\bar{R} = 4.7667\text{ml}$$

(b) Formula: $\bar{\bar{X}} \pm A_2\bar{R}$ w/ $A_2 = 0.577$

$$A_2\bar{R} = 0.577 \times 4.7667$$
$$= 2.750$$

$$UCL_{\bar{X}} = 500.975 + 2.750$$
$$= 503.725$$

$$LCL_{\bar{X}} = 500.975 - 2.750$$
$$= 498.225$$

$UCL = D_4 \bar{R}$ $\quad LCL = D_3 \bar{R}$, $D_4 = 2.114$, $D_3 = 0$

$UCL_R = 2.114 \times 4.7667$ $\quad LCL_R = 0 \times 4.7667$

$UCL_R = 10.077$ $\quad LCL_R = 0$

(c) $\max(\bar{X}) = 503.4 \le 503.725 = UCL_{\bar{x}}$

8th subgroup

$498.225 = LCL_{\bar{x}}$

$\min(\bar{X}) = 498.8 > 498.225$

2nd subgroup

Plot !!!

All 12 pts lie within the limits. Therefore the process is in statistical control.

$\max(R) = 7.1$ (8th) $\qquad \min(R) = 2.8$ (7th)

Question 2

a. $\bar{p}$ (average fraction of defectives $\bar{p}$)

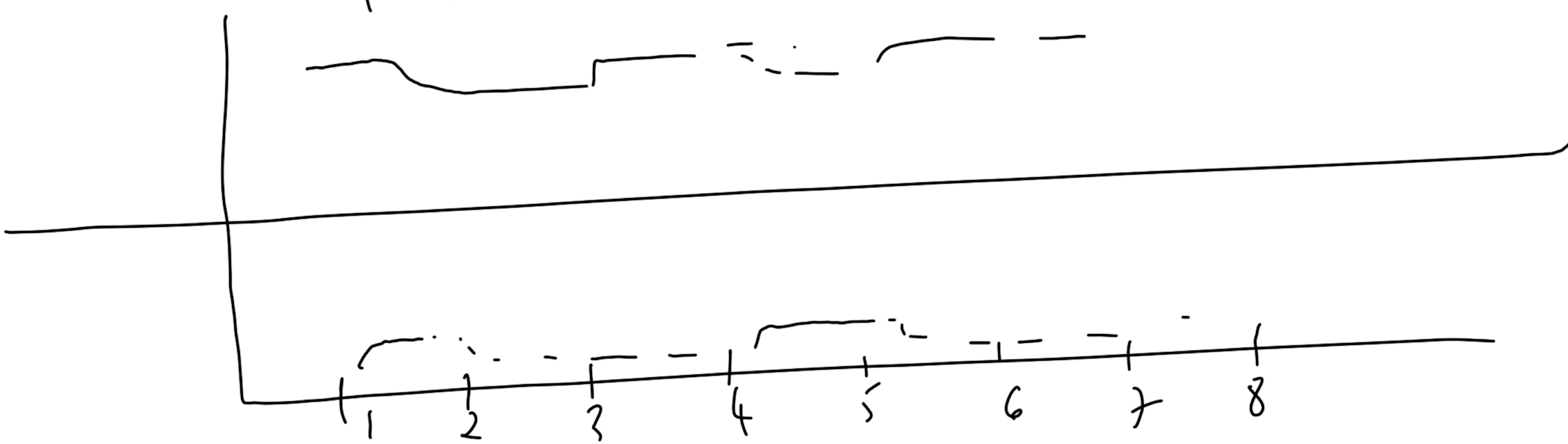
$$\bar{p} = \frac{\sum D_i}{\sum n_i} = \frac{81}{1790} = 0.0453 \quad (4.53\%)$$

b. UCL and LCL for each batch

$$\bar{p} \pm 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$

Batch	n_i	$p_i = D_i/n_i$	UCL	LCL
1	200	0.04	0.0893	0.0012
2	250	0.0480	0.0848	0.0057
3	180	0.0278	0.0918	0
4	220	0.0682	0.0875	0.0030
5	300	0.0300	0.0813	0.0092
6	190	0.0737	0.0905	0
7	240	0.0292	0.0855	0.0050
8	210	0.0524	0.0884	0.0021

$\max_{i=1}^{12}(p_i) =$



All 8 batches fall within their specific batch limit so the defect-producing is in statistical control at about 4.5 % defective level

Question 3.a P122-134

$n = 10$ parts, 3 operators, $r = 2$ replicates, $\bar{R} = 0.068$, $\bar{X}_{diff} = 0.13\,mm$

- Equipment Variation (repeatability) : $\boxed{EV = 5.15 \times \frac{\bar{R}}{d_2}} = 0.3105\,mm$

- Appraiser Variation (reproducibility) : $\boxed{AV = \sqrt{(X_{diff} \times K_2)^2 - \frac{EV^2}{n \times r}}}$

$K_2 = 0.5231$

$$AV = \sqrt{(0.13 \times 0.5231)^2 - \frac{0.3105^2}{10 \times 2}} = \sqrt{-0.000195} \text{ (absurd)}$$

We set $\boxed{AV = 0}\,mm$

- $$\boxed{\text{Gauge R\&R} = \sqrt{EV^2 + AV^2}} = \sqrt{0.3105^2 + 0} = 0.3105 \text{ mm}$$

- $$\boxed{\%GR\&R = \frac{\text{Gauge R\&R}}{TV}}$$; $$\boxed{\underset{\text{Total Variation}}{TV} = \sqrt{(\text{Gauge R\&R})^2 + PV^2}}$$

$$\boxed{\underset{\text{Part Variation}}{PV} = \bar{R}_{part} \times K_3}$$

not provided